



ශ්‍රී ලංකා තාක්ෂණික විශ්වවිද්‍යාලය - පර්යේෂණ

SLTC Research University

**Bachelor of Applied Information Technology
Degree Programme – (BAIT)**



"The battleline between
good and evil runs
through the heart of every
man."
- Aleksandr Solzhenitsyn



Module: **CIT110**

BAIT

Object Oriented Approach to Programming



python™



GUI Programming - Widgets

Slider - Widget

GUI silder - Saved as slider1.py

```
import tkinter as tk
from tkinter import ttk    # ttk stands for "themed tk" - an extension of the tkinter module

def move_slider(event):
    value = slider.get()
    label.config(text=f"Slider Value: {value:.2f}")

root = tk.Tk()             # Create the main window
root.title("Slider Widget")
root.geometry("250x250")

label = ttk.Label(root, text="Slider Value: 0") # Label to display the slider value
label.pack(pady=10)

# Create a slider widget
slider = ttk.Scale(root, from_=0, to =50, orient="horizontal", command= move_slider)
slider.pack(pady=10)

root.mainloop()          # Start the GUI event loop
```

Slider – Widget Application

GUI silder - Saved as slider4.py

```
import tkinter as tk
def update_radius(radius):
    canvas.delete("circle")
    canvas.create_oval(150 - radius, 150 - radius, 150 + radius, 150
                      + radius, outline="red", fill="white", width = 3, tags="circle")
def move_with_slider(event):
    radius = slider.get()
    update_radius(radius)
window = tk.Tk()                                # Create the main window
window.title("Changing the circle Size with the Slider")
canvas = tk.Canvas(window, width=300, height=300) # Canvas to draw the circle
canvas.pack()
# Create a slider widget
slider = tk.Scale(window, from_=0, to=100, orient=tk.HORIZONTAL, command=move_with_slider)
slider.pack()
initial_radius = 50                             # Set a default radius
update_radius(initial_radius)
slider.set(initial_radius)
window.mainloop()                               # Start the main event loop
```

Slider – Widget Application

GUI silder - Saved as vol2.py

```
import tkinter as tk
from pycaw.pycaw import AudioUtilities, ISimpleAudioVolume

def set_volume(volume):
    sessions = AudioUtilities.GetAllSessions()
    for session in sessions:
        volume_object = session._ctl.QueryInterface(ISimpleAudioVolume)
        volume_object.SetMasterVolume(volume, None)

def on_volume_change(value):
    volume = float(value) / 100
    set_volume(volume)

root = tk.Tk()
root.title("Volume Control")

volume_slider = tk.Scale(root, from_=0, to=100, orient=tk.HORIZONTAL,
command=on_volume_change)
volume_slider.set(50)
volume_slider.pack()

root.mainloop()
```

Slider – Widget Application

GUI silder - Saved as vol4.py

```
import tkinter as tk
from pycaw.pycaw import AudioUtilities, ISimpleAudioVolume

def set_volume(volume):
    sessions = AudioUtilities.GetAllSessions()
    for session in sessions:
        volume_object = session._ctl.QueryInterface(ISimpleAudioVolume)
        volume_object.SetMasterVolume(volume, None)

def on_volume_change(value):
    volume = float(value) / 100
    set_volume(volume)

def create_volume_control():
    root = tk.Tk()
    root.title("Volume Control")
    root.geometry("350x200")

    volume_slider = tk.Scale(root, from_=100, to=0, orient=tk.VERTICAL
                             ,length = 200
                             ,font = ('Aerial',12)
                             ,tickinterval = 50
                             ,resolution = 0.5
                             ,label = 'vol'
                             ,troughcolor = 'orange'
                             ,command=on_volume_change)

    volume_slider.set(10)
    volume_slider.pack(pady=10)

    root.mainloop()

if __name__ == "__main__":
    create_volume_control()
```

End of Lesson 10